FLAT AND LOCALLY FINITELY PRESENTED WITH ETALE FIBRES IS ETALE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A fibre which is a disjoint union of spectra of finite separable extensions is etale over the point. Lemma \ref{lemma-etale-over-field}, cited at proof lines 8226-8227. Makes the fibres smooth.
- Flatness with smooth fibres gives smoothness, and the corresponding fibre criterion for being unramified. Lemma \ref{lemma-smooth-flat-smooth-fibres} and Lemma \ref{lemma-unramified-etale-fibres}, cited at proof lines 8228-8231. Give smoothness and unramifiedness.
- A smooth and unramified morphism is etale. Lemma \ref{lemma-etale-smooth-unramified}, cited at proof lines 8232-8233. Concludes.

Source reference. Original lemma 02GM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3233–3233. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00U6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1100–1100. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02GL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3232–3232. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01V8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2437–2437. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02G8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3219–3219. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02GK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3231–3231. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. If f is flat, locally of finite presentation, and for every $s \in S$ the fibre X_s is a disjoint union of spectra of finite separable field extensions of $\kappa(s)$, then f is étale.*

Proof. You can deduce this from Algebra, Lemma 00U6. Here is another proof.

By Lemma 02GL a fibre X_s is étale and hence smooth over s. By Lemma 01V8 we see that $X \to S$ is smooth. By Lemma 02G8 we see that f is unramified. We conclude by Lemma 02GK. □